

sEMG-Based Gesture Classification for Real-Time Embedded Prosthetic Control

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I. ABSTRACT

Abstract- This project presents a EMG classification system for responsive and accurate real-time prosthetic control, which is crucial to address high abandonment rates of modern prosthetics. The classification system detects four hand states—rest, grip, hold, and release—using only a single-channel surface EMG signal, a departure from multi-channel state-of-the-art systems. Data was collected from 18 healthy participants, and five neural network architectures, two ANNs and three LSTMs, were evaluated across three pre-processing pipelines. Model performance was assessed using per-subject cross-validation and macro F1 score, with inference times of the best-accuracy models measured on a BeagleBone Black to assess real-time viability. The Artificial Neural Network (ANN) with the most extensive pre-processing was the best-performing model, achieving a mean F1 score of 0.666 and a total inference time of 273 ms, successfully meeting the critical 300 ms real-time latency requirement for responsive prosthetic control. While good accuracy was demonstrated for distinguishing steady hand states, a key challenge identified was the classification of transient-states (grip and release). This stemmed from a lack of data, class imbalances, labeling inaccuracies, and analog filtering in the acquisition hardware. Future work should focus on mitigating these issues by capturing an expanded, balanced dataset with unfiltered EMG data to enhance transient-state classification.

Index Terms—Classification, EMG, Embedded System, Myoelectric Control, Neural Network

II. INTRODUCTION

The loss of a limb can significantly impair an individual's ability to perform everyday tasks and interact with their environment. Modern prosthetic technologies aim to restore some of this lost functionality by enabling intuitive and responsive control of artificial limbs [1]. Among various control strategies, electromyography (EMG) based control has emerged as a promising approach due to its ability to capture neuromuscular activity that directly reflects motor intent [2]. These signals can be processed and classified using AI techniques to infer the

user's intended motion.

Despite advancements in modern prosthetics, a significant number of users end up abandoning their prosthetic devices due to issues such as discomfort and poor performance (slow and inaccurate responses) [3]. It is therefore important that any prosthetic device is compact, responsive, and accurate for long-term adoption. For effective responsive real-time control, total system latency must remain below 300 ms to avoid user-perceived lag [4]. These requirements impose real constraints on the computational resources of modern prosthetics. A compact form factor limits the size of the embedded hardware, including the processing unit, while the need for responsiveness requires that classification operates in real-time [2], [5].

Given the hardware and time constraints, the model's size and complexity are limited, which poses challenges to achieve high classification accuracy. Furthermore, EMG signals are inherently noisy, which further increases the challenge of classification [6]. This report presents the design and implementation of an EMG-based classification systems for controlling a four gesture hand prosthetic in real-time. The study covers data acquisition, pre-processing, feature extraction, classification using AI models, and deployment on an embedded device.

A. Background

EMG sensors detect the electrical activity generated by skeletal muscles. There are two types of EMG sensors, internal- and surface- EMGs. In biomedical research the most commonly used EMG sensor type is surface EMG (sEMG) due to their ease of use [2]. sEMG sensors are prone to noise from the environment, motion artifacts (subjects moves other parts of their body), and muscle crosstalk.

To remove this noise from the measured sEMG data the signals are pre-processed. Common pre-processing methods includes bandpass filtering (which removes low frequency motion artifacts as well as high frequency sensor noise), notch filtering (which removes power line interference at 50 Hz), and normalization to ensure standardized value ranges [7]. Other possible pre-processing methods include the moving average operation (which helps further reduce noise by smoothing the signal), and the absolute value operation (which enables the classifier to consider only the magnitude of the signal) [8].

Once an EMG signal has been pre-processed, feature extraction can be applied to reduce the dimensionality of the raw data and extract new information from the data. Feature extraction is typically done in one of three domains:

- **Time domain (TD):** Methods such as mean absolute value (MAV), variance (VAR) [9], and Zero Crossing (ZC) [2].
- **Frequency domain (FD):** Methods such as the Fourier transform (FT) which decomposes a signal containing multiple frequencies into the signals frequency elements along with their magnitudes [10], [11].
- **Time-frequency domain (TFD):** Methods such as short-time Fourier transform (STFT) which is a type of FT that is applied multiple times over the length of a signal enabling analysis of both the frequency and temporal characteristics of the signal [9], [11].

After feature classification the EMG signal can be classified. The classification of EMG signals is typically done using either machine learning, deep learning, or a combination. Traditional machine learning models, such as K-Nearest Neighbor (KNN), can provide high accuracy, but require a large amount of manual feature engineering to achieve it. Deep learning methods, such as Artificial Neural Network (ANN) and Long Short-Term Memory (LSTM) instead let the model learn how to extract usable features directly from the signal. According to Zaim et al. [12] deep learning models show high accuracy and are the most popular in recent research compared to machine learning solutions, despite the limitations posed by computational and real-time constraints inherent in prosthetics [5], [2].

B. Related Work

Several studies have explored EMG-based control systems for hand prostheses using a variety of classification techniques. A study by Ekrad et al. [13] utilized one sEMG channel to classify 10 individual finger states (open and close motion), designed for real-time control of a robotic hand. The study implemented two deep learning methods, an ANN and a Recurrent Neural Network (RNN). The RNN achieved the best accuracy of 74%, while the ANN reached an accuracy of 61%. No real-time performance was presented in the report. The study used only 3 subjects to gather data, which resulted in a limited dataset.

Another study by Hye et al. [9] explored different methods to recognize 53 hand movements from 10 sEMG channels to control a prosthetic hand. This study investigated multiple learning models and they found the best performance from a KNN and an ANN. They found that the KNN had an accuracy of 97.5% while the ANN had a slightly higher accuracy of 97.7%. The study did not publish inference speed, but comparing the required computation time the ANN required 43 000 seconds of computation to achieve the same accuracy as the KNN, which only required 2 seconds of computation time. The study used 27 subjects, which provided a much more diverse dataset as compared to the study by Ekrad et al. [13].

A third study by Toro-Ossaba et al. [8] explored EMG-based hand gesture recognition using an LSTM combined with dense layers to classify five hand gestures in real time using only four EMG channels. Their model achieved 99% accuracy during training and validation, and $87\% \pm 7\%$ accuracy in real-world testing. This study showed that a compact LSTM model can perform well on few EMG channels while being suitable for use on an embedded systems. The study used 8 healthy subjects with an age range between 20 and 30 years.

C. Scope and Research Questions

This project aims to classify four hand states (rest, grip, hold, and release) using a single sEMG channel. Optimal electrode placement will first be determined to capture the most informative signal. Data will be collected into a dataset which is then processed through multiple pre-processing pipelines. Models will be trained and evaluated on this dataset. Finally, the best performing models will then be deployed on an embedded system to assess real-time performance.

The study is limited by the use of data from only healthy subjects, which may reduce generalizability. Real-time constraints on embedded hardware impose limits on model complexity, memory use, and latency. Additionally, the limited number of EMG channels restricts spatial resolution, potentially affecting classification accuracy. The project investigates how classification models perform under these constraints, guided by the following research questions:

- RQ1: Which of the five tested AI classification model architecture achieves the highest macro F1 score while meeting real-time inference latency constraints for single-channel sEMG hand state recognition when deployed on an embedded system?
- RQ2: How does classification performance vary across the four target hand states for the best performing AI models?
- RQ3: How does the interaction between sEMG signal pre-processing degree (minimal to extensive) and time-frequency domain feature extraction methods (short-time Fourier transform) influence classification accuracy?

III. METHOD

This section describes the methodology for a real-time, single-channel EMG system classifying four hand states. The method involves: (1) Standardized data acquisition using sEMG sensors; (2) Pre-processing pipelines; (3) Classification using compact ANN/LSTM architectures; (4) Feature extraction in the time domain or the time-frequency domain; (5) Validation using per-subject cross-validation and macro F1 score.

A. Data acquisition

Accurate and reproducible acquisition of sEMG signals requires standardized protocols for electrode placement, signal labeling, and task design. Guidelines from the SENIAM project [14], Delsys Technical NOTE 101 [15], and the recommendations by Zipp [16] provide detailed procedures for these aspects. These references emphasize aligning electrodes with

muscle fibers, avoiding tendons, and maintaining consistent electrode spacing to minimize signal artifacts and crosstalk.

The choice of which muscle to measure significantly impacts both the signal quality and task relevance [16]. Although the Flexor Carpi Radialis and Extensor Carpi Radialis Longus are frequently used in forearm studies, they may show limited activation during simple grip movements. The Flexor Digitorum Superficialis (FDS), by contrast, is directly involved in finger flexion and is recommended for EMG recordings due to its anatomical accessibility and reliable activation during grasping tasks.

This study placed sensors on the FDS, following anatomical recommendations from SENIAM and Delsys, meaning the spacing between the active electrodes was at 2-3 cm and the electrodes were placed on the muscle belly while avoiding tendon areas. A sampling rate above 1000 Hz is often recommended for EMG acquisition, as it exceeds the Nyquist limit for typical muscle signal frequencies which is 20–500 Hz [6], [17]. Increasing sampling rates further minimizes aliasing while supporting more advanced pre-processing techniques.

Signal labeling in EMG classification is often done using external cues like button presses or videos to align phases (rest, movement, hold) with timestamps [13]. This method improves timing accuracy and reduces the need for motion tracking. In this study, a dynamic script automatically labeled signals based on these cues.

B. Pre-Processing

While standardized acquisition protocols ensure signal integrity, raw sEMG data contain artifacts and noise. Due to this, the collected signals need to be pre-processed to ensure the quality of the signal for subsequent feature extraction and classification stages.

The signal pre-processing in this work follows the methodology described by Toro-Ossaba et al. [8]. Their study demonstrated that mean subtraction, absolute value transformation, moving average filtering, and normalization are viable methods for improving the quality of the EMG signal while being suitable for real-time use on embedded systems. In addition to these techniques a bandpass filter was applied to remove both low-frequency motion artifacts and high-frequency noise, preserving the frequency components most relevant to the measured muscle activity. Notch filters, which often are used to eliminate power line interference, was intentionally left out to avoid attenuating valuable signal components. Segmentation of the signals were done using an overlapping window. This was chosen over an adjacent window, to ensure that transition events were not missed.

C. Classification

This study explored three primary approaches for EMG signal classification: k-Nearest Neighbors (KNN), Artificial Neural Networks (ANN), and Long Short-Term Memory (LSTM). The selection of algorithms was informed by literature but decisions were made based on the the constraints of single-channel EMG data and limited sample size.

In a study by Hye et al. [9], KNN achieved an accuracy of 97.5%. Despite the high accuracy, KNN was excluded on the basis that it requires 10 EMG channels to achieve the performance while this study uses only a single EMG channel.

The same study by Hye et al. [9] showed that an ANN using a 5-layer ANN with 1000+ units per layer and the same 10-channel EMG system can achieve an accuracy of 94%. However, the large amount of training data required to train such a large ANN [18] is beyond what is available in this study.

Finally, a study by Toro-Ossaba et al. [8] demonstrated that an LSTM can achieve an accuracy of 87%. This was achieved while using only 4 EMG channels with a compact network architecture (32-unit dense layer, 16-unit LSTM layer, final 32-unit dense layer). This compact architecture reduces the computational requirements for real-time inference as well as enables improved generalization potential for a smaller dataset [18].

Initial experiments in this study showed that a compact ANN, similar to the compact LSTM presented by Toro-Ossaba et al., with few layers and units, could provide good accuracy and real-time performance. This suggests that for single-channel EMG data, small feedforward networks and LSTMs both represent effective and viable solutions for real-time motion intention classification on a single-channel EMG.

Based on the presented literature and initial experimentation, this study will investigate both compact LSTMs and compact ANNs as classification solutions.

D. Feature Extraction

Having determined the classifier to be either an LSTM or an ANN, the feature extraction methodology can now be determined. As discussed, feature extraction can be done in one of three domains: the time domain (TD), the frequency domain (FD), and the time-frequency domain (TFD).

Toro-Ossaba et al. [8] showed that an LSTM can learn hand motion recognition on TD EMG data. However, the TD does not contain any information about the frequency characteristics of the signal.

Both the FD and the TFD reveal information about the signal's frequency characteristics, such as the frequency elements and their magnitudes. The frequency elements and their magnitudes contain information which is useful for classifying the motion intention [11]. Of these two methods, TFD preserves the temporal information while also revealing the frequency characteristics of the signal.

This study will compare the effect of training an LSTM and ANN on TD and TFD EMG data. To map the data from the TD to the TFD, the Short-Time Fourier Transform (STFT) is applied. STFT has shown high accuracy as a feature extraction method in classifying motion intention when paired with a Convolution Neural Network (CNN) [19].

E. Validation Metric

Selection of the validation metric is a crucial part of evaluating the model fairly and accurately. This study deals with a dataset

with imbalanced number of examples per class. Due to this, a validation metric that allows for fair and clear comparison in model performance across imbalanced classes is needed.

Three evaluation metrics were considered: accuracy, confusion matrices, and F1 score. Accuracy was rejected due to its unreliability with imbalanced datasets [20]. While confusion matrices provide detailed class-wise insights, their multi-dimensional nature complicates direct model comparisons [20]. The macro F1 score was selected as the primary metric because it mitigates the effect of class imbalances by applying the harmonic mean between precision and recall across all classes. The use of macro F1 score aligns with established best practices for EMG classification [21].

To further increase the reliability of the calculated macro F1 score this study incorporates per-subject cross-validation, where the model is trained on data from all but one subject, the so-called hold-out subject. The macro F1 score is then calculated based on the performance on this hold-out subject, which eliminates data leakage, providing a fair and comprehensive view of each models performance.

The per-subject cross-validation will also provide insight into any potential subject outliers. Any subject more than 2 standard deviations away from the mean F1 score for that model is considered an outlier and is therefore eliminated.

This application of per-subject cross-validation and F1 score allows for a comprehensive evaluation of model performance. The evaluation method effectively addresses any class imbalances in the data as well as prevents any data leakage, both of which can artificially increase the reported performance.

IV. EXPERIMENTATION

This section details the experimental implementation of the methodology, including data acquisition protocols, pre-processing pipelines, model architectures, and embedded deployment procedures.

A. Data acquisition

Each recording session began with a standardized skin preparation to ensure low impedance and signal stability. Participants washed their right forearm with soap and water and then disinfected it with alcohol wipes.

Electrodes were applied over the FDS as described in Method III. A reference electrode was placed on the Medial Epicondyle of the elbow. The final electrode placement layout is shown in Figure 1. During pilot testing, attempts were made to record from the Flexor Carpi Radialis and Extensor Carpi Radialis Longus muscles. However, these muscles showed insufficient signal strength during grasping. While these tests were not formally included in the dataset, they strengthened the decision to use the FDS as the primary target muscle.

Data was collected for two different levels of gripping strength, firm and soft. For collection of the firm grip, a cylinder with an 8 cm diameter and an integrated force sensor was used. Figure 2 shows the grip and release positions with the cylindrical object. For collection of the soft grip, a whiteboard marker was used.

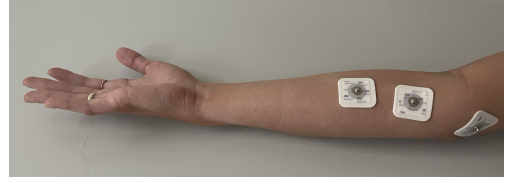


Figure 1. Electrode placement on the right forearm targeting the Flexor Digitorum Superficialis (FDS).



Figure 2. Participant holding the standardized cylindrical object. Left and right picture is rest/release and grip/hold position, respectively. Red light indicates sufficient grip strength is applied.

Participants were instructed to sit still, rest their forearm and elbow in a natural position on the table during all recordings. The room was dimmed to reduce ambient noise, and a 30-second pause preceded each recording session to stabilize the signal. Signs of fatigue were noted when present.

Each participant performed five full cycles. Each cycle consisted of two repetitions of grip-hold-release: one short (3.5 seconds) hold, and one long (9.5 seconds) hold. The total duration of one cycle was approximately 34 seconds. A fixed LED countdown was used to guide the timing. To improve timing precision, a brief beep sound was used to guide participants to grip and release at the correct moments, ensuring alignment with the intended labeling windows.

Signals were recorded using two different signal conditioning modules, the SparkFun AD8232 single-lead ECG/EMG front-end, and the Olimex SHIELD EKG-EMG module connected to an Arduino Uno [22], [23], [24]. Both acquisition systems included built-in analog bandpass filtering. Although these filters were not part of the original design plan, they were accepted for consistency across the dataset. Twisted-pair cables were used in both setups to minimize electromagnetic interference.

A total of nine participants were recorded using the AD8232 module with firm grip of the cylindrical object (five cycles per participant). Another nine participants were recorded using the Olimex + Arduino Uno setup, completing five cycles of firm grip of the cylindrical object and five additional cycles of soft grip of whiteboard marker.

All signals were sampled at 5000 Hz to capture detailed muscle activity and labeling was done automatically based on the LED timings. Due to minor timing offsets from participant reaction times and transition overlap, labels were manually corrected when needed. Recordings with obvious outliers, such as large spikes during rest periods, were manually removed from the dataset.

B. Pre-processing

All EMG signals were pre-processed to reduce noise and improve signal quality. A 7th order Butterworth bandpass filter was applied to the signal with a frequency band of 20-500 Hz. For segmentation, an overlapping window of 200 ms with 25% overlap was used. During pre-processing the collected data was down sampled to 1000 Hz.

To investigate the impact of different pre-processing strategies on model performance, three different pipelines were tested:

- **Pipeline A:** The same pre-processing as described in Toro-Ossaba et al. [8]. The signal is first centered by subtracting the mean, transformed to the absolute value, applying a moving average filter with a window length of 20 samples, and normalized using z-score.
- **Pipeline B:** This pipeline includes mean subtraction, absolute value transformation, and normalization, but omits the moving average filter. It is used to assess the impact of the moving average filter on model accuracy.
- **Pipeline C:** In this pipeline, only mean subtraction and normalization are applied. This pipeline shows the model performance with minimal signal manipulation.

C. Model Training and Validation

This study investigated the performance of five model architectures. All five model architectures are shown in Figure 3.

Architecture (1), LSTM, is based on the study by Toro-Ossaba et al. [8]. However, the initial dense layer used by Toro-Ossaba et al. has been replaced by a 1D convolutional layer in this work. This replacement offers two main advantages. First, the 1D convolutional layer compresses the data, which allows for a reduction in the number of iterations performed by the LSTM layer, thereby providing a significant speedup in both training and inference time. Second, unlike the dense layer, which can only analyze one sample at a time, the 1D convolutional layer can examine multiple, adjacent samples simultaneously.

Architecture (2), LSTM_STFT, and architecture (3), LSTM_STFT_Dense, replaces the initial 1D convolutional layer found in LSTM with the Short-Time Fourier Transform (STFT). The STFT acts as a compression layer, same as the replaced convolutional layer, while also revealing the frequency components and their magnitude found in the data. LSTM_STFT_Dense also incorporates an additional dense layer, analogous to how Toro-Ossaba et al. did in their architecture.

Architecture (4), ANN, and architecture (5), ANN_STFT, are both small ANNs, replacing the LSTM layer with a fully connected layer. This removes the need for recurrent computation, providing speed up in training and inference time. Just as LSTM_STFT and LSTM_STFT_Dense did, the model ANN_STFT, adds an initial STFT layer.

Each of the five model architectures shown in Figure 3 was trained using each of the three pre-processing pipelines, resulting in a total of 15 unique model and pre-processing combinations and per-subject cross-validation was performed

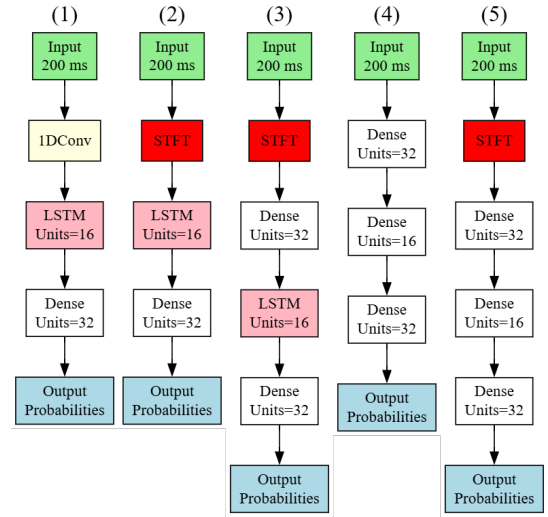


Figure 3. Overview of the five neural network architectures evaluated in this study.

for each configuration. This approach provides a comprehensive evaluation of model performance across subjects and helps identify any subject-specific outliers in the dataset.

D. Deployment

The best trained variant of each model was then implemented on a revision C BeagleBone Black [25]. Before deployment on the embedded device the model is first converted to a TensorFlow Lite 2.19.0 compatible format, a library used for efficient deployment on a resource-constrained devices. Each pre-processing pipeline was deployed on the BeagleBone as well as the windowing logic.

To measure the real-time performance EMG data was collected in real time using sensors connected to the embedded device. The collected data was then segmented, pre-processed and fed to a classification model. Multiple runs per configuration ensured robust and fair measurements of mean inference time and total response time.

V. RESULTS

This study investigated three pre-processing pipelines and five model architectures, two non-STFT models and three STFT models.

Analysis of the results after outlier removal, shown in Figure 4, reveal that Artificial Neural Networks (ANNs) show the overall best performance. The single best model is ANN combined with pre-processor pipeline A, achieving a mean F1 score of 0.666 on the per-subject cross-validation. The second and third best models are ANN with pre-processor B (mean F1 0.654) and ANN_STFT with pre-processor C (mean F1 0.644).

Following segmentation of the dataset, a total of 11,934 labeled examples were obtained. The label distribution consisted of the steady states, rest and hold, which accounted for 56.9% and 40.5% of the dataset, respectively. The transient states, grip and release, each comprised 1.3% of the total dataset. This distribution is reflected in the average per-class F1 score

for the top 3 models, as shown in Figure 5. The steady states rest and hold achieved an average F1 score of 0.98 and 0.96 respectively, while transient states grip and release attained an average F1 score of 0.38 and 0.23 respectively.

Evaluation of the pre-processor performance can be seen in Table I. This table shows that for both STFT models and for non-STFT models pipeline A attains the best mean F1 score. The table also shows that overall, more pre-processing achieves a higher F1 score than less pre-processing. However, for STFT models, this pattern does not appear, showing less variation in performance based on pre-processing pipeline.

Also shown in Table I, is the performance of STFT vs non-STFT models. STFT models shows overall higher performance (overall mean F1 of 0.566 for STFT compared to 0.549 for non-STFT), but non-STFT has the highest peak performance (peak mean F1 of 0.644 for non-STFT compared to 0.589 for STFT).

Table II shows the inference time on the Beaglebone Black for the five models with highest F1 scores as well as the processing time for the associated pipeline. The table does not include the time to collect a 200 ms window with 25% overlap. Of these five models one model, LSTM_STFT, was unable to run on the BeagleBone Black.

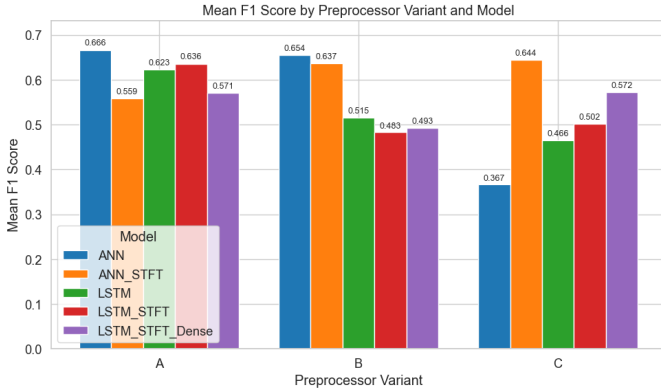


Figure 4. Mean F1 scores from per-subject cross-validation for each of the 15 unique model and pre-processing combinations evaluated in this study.



Figure 5. Class-wise F1 scores for the top three performing models.

Table I
MEAN F1 SCORES OF STFT AND NON-STFT MODELS ACROSS PRE-PROCESSING VARIANTS

Pre-processor	STFT Models Mean F1	non-STFT Models Mean F1	Overall Mean F1
Overall	0.566	0.549	0.559
(A)	0.589	0.644	0.611
(B)	0.538	0.585	0.557
(C)	0.573	0.416	0.510

Table II
INFERENCE TIME PER BEST MODEL VARIANT ON THE BEAGLEBONE BLACK

Models	Pipeline	Mean Processing	Mean Inference	Mean Total Time
ANN	A	46 ms	77 ms	123 ms
ANN_STFT	C	44 ms	80 ms	124 ms
LSTM	A	45 ms	130 ms	175 ms
LSTM_STFT	A	N/A	N/A	N/A
LSTM_STFT_Dense	C	45 ms	86 ms	131 ms

VI. DISCUSSION

The experimental results demonstrate the potential of single-channel sEMG based classification of hand states. Among the evaluated models, the Artificial Neural Network ANN emerged as the most effective architecture, showing both the highest classification accuracy (mean F1 = 0.666) as well as the fastest inference time (total inference time = 123 ms). The total inference time in Table II does not include the time to collect a 200 ms window with 25% overlap. This means the true inference time is $123 + 150 = 273$ ms, which is below the 300 ms real-time requirement used in this study.

To understand the impact a Short-Time Fourier Transform (STFT) layer has on model inference time a comparison between the models ANN and ANN_STFT is done. The only architectural difference between these models is the addition of the STFT layer. Here we find that the inference time increases by only 5 ms, or 3.2% of the total inference. Furthermore, the larger LSTM_STFT_Dense model achieved a lower mean total inference time of 281 ms (when including windowing time) when compared to the LSTM model with a mean inference time of 325 ms (which means LSTM model does not meet real-time requirements). This is despite LSTM_STFT_Dense having one more layer than LSTM, further emphasizing the efficiency of the STFT layer. The mean performance of the models (mean F1 for LSTM = 0.534; mean F1 for LSTM_STFT_Dense = 0.545) shows that LSTM_STFT_Dense also has a marginally better overall accuracy. This suggests that the STFT layer is more efficient at reducing data dimensionality and extracting features than a 1D convolutional layer for single EMG channel data.

The addition of an STFT layer only increases inference time by 5 ms while making the model more robust (less variation in F1 score) to the degree of pre-processing. However, non-STFT models F1 scores were affected greatly by the degree of pre-processing, where more extensive processing shows better F1 score. The variation in performance within each pre-processor

group (especially for non-STFT models) suggests that there is no one-size fits all when it comes to pre-processing. This suggests that the pre-processing needs to be tailored to the specific classification model. It is also worth noting that, while non-STFT models have slightly lower total mean F1 scores than STFT models, they show greater peak performance.

While the effect of pre-processing degree had a noticeable impact on the performance of the models, only minor differences in processing times were observed across the three pre-processing pipelines. As shown in Table II, the difference between the most computationally intensive pipeline (A, 45-46 ms) and the lightest (C, 44-45 ms) was only 1-2 ms, or about 1-2% of the total processing time. This suggests that the 7th order Butterworth bandpass filter, which all models used and is known to be computationally intensive, likely contributed the most to the pre-processing time, while the remaining pre-processing steps contributed negligibly to the inference latency.

Finally, even the best performing models struggle to accurately classify the transient states (grip and release) in contrast to the steady states (rest and hold). Based on the results of this study, this can be due to the following five factors: (1) the limited size of the dataset, (2) class imbalance in the dataset, (3) inaccuracies in the manual labeling near the transition boundaries, (4) built-in filtering applied during the data acquisition, and (5) the limited spatial resolution inherent in single channel sEMG. In particular, the severe underrepresentation of grip and release (each constituting only 1.3% of the dataset) was identified as the main contributor to the models difficulty in learning effective representation of transient states.

A. Future Work

As stated, the biggest challenge in this study was classifying transient-states. In the above section multiple theories for why classifying transient states is such a challenge is laid out, focusing mainly on data acquisition.

Any future work in single-channel EMG hand state classification should focus on improving the data availability and quality. Data availability could be improved by collecting higher number of samples from test subjects or by investigating the use of simulated data. Data quality could be improved through the use of multiple methodologies. Using an unfiltered data acquisition device could yield more expressive data, potentially aiding the classifier in classifying transient-states. Improving labeling of the samples, including more precise timing and better transition definition between states, could also improve the final classification performance. The final data quality improvement could be to collect data in such a way as to minimize the class imbalances, especially those found between steady- and transient-states.

VII. CONCLUSION

A key part in creating accessible and functional prosthetics is responsive and accurate classification of the user's motion intention. This study investigated the classification of four

hand states (rest, grip, hold and release) using a single-channel sEMG system deployed on an embedded device. Three pre-processing pipelines (most to least extensive) and five models (two non-STFT models and three STFT models) were compared. This study found that a non-STFT ANN with extensive pre-processing achieved the highest classification F1 score and the lowest inference time. However, no systematic difference between STFT and non-STFT models was found, though STFT-models were shown to be more robust to variations in the degree of pre-processing. The classification accuracy showed large variation across hand states: steady-states (rest and hold) were reliably classified, while transient-states (grip and release) were not. This study theorized that these issues stemmed from limited data availability, class imbalances, labeling issues, and analog filtering during acquisition. Future work should prioritize: larger, balanced datasets; improved labeling; and unfiltered signal acquisition which will aid in capturing transient-state features. These steps will help bridge the gap between laboratory validation and real-world prosthetic control.

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